

SIMATS ENGINEERING
ASSESSMENT TOOL 1-INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1-INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
Student Name:	KV GAGAN YADAV	Reg.No.:	192412406
Department:	ECE	Date:	20/08/26

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

- Smartphone Performance Optimization**
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
- Cloud Server Memory Bottleneck**
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
- Autonomous Vehicle Data Processing**
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
- AI Inference Edge Device**
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
- High-Performance Gaming Console**
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

***I KV GAGAN YADAV (192412406) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.***

Signature of the student:

KV GAGAN YADAV

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Q1. Smartphone Performance Optimization

Problem Statement:

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

1. Understanding of the Industry Problem

When a user switches from one app to another, the processor needs to quickly access the instructions and data of the new application. If the required data is not available in the cache, the processor has to access DRAM.

This creates **cache misses and memory delays**, which can be experienced by the user as lag or stuttering.

Therefore, the main problem is to **reduce memory access latency and improve memory bandwidth** during application switching.

2. Role of Memory Hierarchy Elements

L1 Cache:

- It is the smallest and fastest cache.
- It is located very close to the CPU core.
- It stores frequently used instructions and data.
- It provides very low access latency.

L2 Cache:

- It is larger than L1 but slightly slower.
- It stores data that cannot fit in L1.
- It reduces the number of accesses to main memory.

L3 Cache:

- It is larger than L1 and L2.
- It is generally shared among processor cores.
- It can store data from multiple applications.
- It reduces expensive DRAM accesses.

DRAM:

- It is the main memory of the smartphone.
- It has much larger capacity than cache.
- It is slower than L1, L2 and L3.
- It stores the working data of running applications.

3. Application of Engineering Concepts

The important concept used here is **locality of reference**.

- **Temporal locality:** Recently used data is likely to be used again.
- **Spatial locality:** Data near recently accessed data is likely to be accessed soon.
- **Cache associativity:** Higher associativity can reduce conflict misses.

- **Prefetching:** Frequently required data can be loaded into cache before it is needed.
- **Cache partitioning:** A portion of cache can be reserved for the foreground application.

4. Proposed Hierarchical Memory Redesign

The proposed hierarchy is:

CPU → L1 Cache → L2 Cache → L3 Cache → DRAM → Storage

I propose the following improvements:

1. Increase the effective **L2 cache capacity** to reduce cache misses.
2. Use a larger **shared L3 cache** to retain useful application data.
3. Use **cache partitioning** to give priority to the foreground application.
4. Use **prefetching** for applications that are likely to be opened next.
5. Use high-bandwidth **DRAM** to reduce the penalty of cache misses.
6. Use an efficient cache replacement policy to retain frequently used data.

5. Use of Tools/Technologies

The redesigned memory system can be tested using:

- **gem5** – memory and processor-system simulation.
- **CACTI** – cache timing, area and power analysis.
- **Perfetto** – performance analysis and profiling on Android systems.
- **High-bandwidth DRAM** – improves data transfer performance.

6. Analysis / Expected Improvement

Assume:

- L3 hit rate before redesign = **70%**
- L3 latency = **35 cycles**
- DRAM latency = **200 cycles**

Before redesign:

$$\text{AMAT} = (0.70 \times 35) + (0.30 \times 200)$$

$$= 24.5 + 60$$

$$= \mathbf{84.5 \text{ cycles}}$$

After cache partitioning and prefetching, assume L3 hit rate improves to **88%**.

After redesign:

$$\text{AMAT} = (0.88 \times 35) + (0.12 \times 200)$$

$$= 30.8 + 24$$

$$= \mathbf{54.8 \text{ cycles}}$$

Therefore,

Percentage reduction:

$$\approx \mathbf{35.1\%}$$

Thus, the proposed design can significantly reduce average memory access latency.

7. Conclusion

The smartphone lag is mainly caused by frequent cache misses and slow DRAM accesses during application switching. By improving the L2 and L3 cache, using cache partitioning, prefetching and high-bandwidth DRAM, the memory hierarchy can provide **lower latency and higher throughput**. This results in smoother and faster app switching.

Q2. Cloud Server Memory Bottleneck

Problem Statement:

A cloud service provider experiences high latency in database queries. Compare cache hierarchy with virtual memory paging and recommend improvements using appropriate memory technologies.

1. Understanding of the Industry Problem

In a cloud server, database applications continuously access records, indexes and instructions. If the required information is not available in fast memory, the processor has to access slower memory.

High database-query latency can therefore occur because of **cache misses, insufficient DRAM and excessive virtual-memory paging**.

2. Cache Hierarchy

The cache hierarchy consists of:

- **L1 Cache:** Fastest and smallest cache.
- **L2 Cache:** Larger than L1 and provides the next level of fast access.
- **L3 Cache:** Larger shared cache that reduces DRAM accesses.
- **DRAM:** Large main memory used to hold active database data.

The hierarchy is:

CPU → L1 → L2 → L3 → DRAM

The main advantage is **low latency**.

3. Virtual Memory Paging

Virtual memory allows programs to use a larger logical memory space than the available physical RAM.

Memory is divided into **pages**. If a required page is not present in RAM, a **page fault** occurs and the operating system loads the page from secondary storage.

Paging provides flexibility and memory protection, but storage access is much slower than cache or DRAM.

4. Comparison

Feature	Cache Hierarchy	Virtual Memory Paging
Main purpose	Reduce CPU access latency	Extend logical memory
Levels	L1, L2, L3	Pages in RAM/storage
Speed	Very high	Lower

Feature	Cache Hierarchy	Virtual Memory Paging
Storage	SRAM/cache	DRAM + secondary storage
Main problem	Cache miss	Page fault
Best use	Frequently accessed data	Large memory requirements

5. Proposed Solution

The proposed hierarchy is:

CPU → L1 → L2 → L3 → High-capacity DRAM → NVMe SSD

Improvements:

1. Increase DRAM capacity so the database working set stays in RAM.
2. Use a larger L3 cache.
3. Use database buffer pools for frequently accessed records.
4. Reduce unnecessary paging.
5. Use fast NVMe SSD for virtual-memory backing storage.
6. Monitor memory usage and page-fault rate.

6. Analysis

If more database data remains in DRAM, the number of storage accesses decreases.

Fewer cache misses + fewer page faults = lower query latency.

Therefore, sufficient DRAM should be the first priority, followed by efficient cache utilization and fast secondary storage.

7. Conclusion

Cache hierarchy mainly reduces **CPU access latency**, while virtual memory provides a larger logical memory space. For cloud database servers, a combination of **large DRAM, efficient L1/L2/L3 caching and fast NVMe storage** provides better performance and lower query latency.

Q3. Autonomous Vehicle Data Processing

Problem Statement:

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

1. Understanding of the Industry Problem

An autonomous vehicle receives continuous data from:

- Cameras
- LiDAR
- Radar
- Ultrasonic sensors
- GPS and other sensors

This data must be processed quickly. Any unnecessary memory delay can affect real-time decision making.

Therefore, the memory system requires **low latency, high bandwidth and sufficient capacity**.

2. Comparison of SRAM, DRAM and MRAM

SRAM:

- Very fast.
- Low latency.
- High cost.
- Lower storage density.
- Suitable for cache and small buffers.

DRAM:

- Larger capacity.
- Lower cost per bit than SRAM.
- Higher latency than SRAM.
- Suitable for large sensor datasets and AI processing.

MRAM:

- Non-volatile memory.
- Retains data without power.
- Good endurance.
- Can be used for firmware, configuration and selected frequently accessed data.

3. Proposed Memory Hierarchy

The proposed structure is:

Sensors → SRAM/Cache → L1/L2/L3 → High-bandwidth DRAM → MRAM

4. Design

1. Store time-critical sensor data in SRAM.
2. Use L1/L2/L3 caches for frequently accessed processing data.
3. Use high-bandwidth DRAM for large sensor frames and AI intermediate data.
4. Use MRAM for non-volatile firmware and configuration.
5. Use multiple memory channels to improve bandwidth.
6. Use DMA to transfer sensor data efficiently.

5. Analysis

SRAM provides **minimum latency**, while DRAM provides **large capacity and high bandwidth**.

MRAM provides **non-volatile storage**, which is useful for important persistent information.

Thus:

SRAM → fastest access

DRAM → large working memory

MRAM → persistent data

6. Conclusion

A combination of SRAM, DRAM and MRAM provides a balanced memory hierarchy for autonomous vehicles. SRAM handles time-critical operations, DRAM handles large sensor data and MRAM stores important non-volatile information. This architecture helps maximize bandwidth while minimizing latency.

Q4. AI Inference Edge Device

Problem Statement:

An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM and RRAM and recommend an optimized hierarchical memory structure.

1. Understanding of the Industry Problem

An edge AI device needs to load an AI model and perform inference locally. The model may contain a large number of parameters.

The challenge is to achieve:

- Fast model loading
- Low power consumption
- Sufficient storage capacity
- Fast inference

2. Comparison of Memory Technologies

NAND Flash:

- Non-volatile.
- High storage capacity.
- Low cost.
- Used for permanent model and operating-system storage.
- Slower than DRAM.

DRAM:

- Volatile.
- Faster than NAND Flash.
- Used for active AI models and intermediate data.
- Requires continuous power.

RRAM:

- Non-volatile.
- Can provide fast access.
- Potentially low-power.
- Useful for selected AI memory applications.

3. Proposed Hierarchy

AI Processor → SRAM/Cache → DRAM → RRAM/NAND Flash

4. Design

1. Store frequently accessed data in fast SRAM/cache.
2. Load the AI model from NAND Flash into DRAM.
3. Keep active model parameters and intermediate results in DRAM.
4. Use RRAM for suitable frequently accessed non-volatile data.
5. Apply model compression to reduce memory requirements.
6. Use quantization to reduce model size and memory traffic.
7. Use low-power memory modes when the device is idle.

5. Analysis

NAND Flash provides **high capacity**.

DRAM provides **fast runtime access**.

RRAM can provide **fast non-volatile access with low power**.

Therefore, each technology is used according to its strength.

6. Conclusion

The optimized hierarchy uses NAND Flash for permanent storage, DRAM for active AI processing and RRAM for selected low-power non-volatile operations. This reduces model-loading time while maintaining low power consumption.

Q5. High-Performance Gaming Console

Problem Statement:

A gaming console suffers frame drops during large scene loading. Analyze L3 cache versus DRAM throughput and propose memory hierarchy enhancements.

1. Understanding of the Industry Problem

Modern games contain large:

- Textures
- 3D models
- Maps
- Audio files
- Animation data

When a large scene is loaded, huge amounts of data must move between storage, DRAM, CPU and GPU.

If the data is not supplied quickly enough, the GPU may wait, resulting in **frame drops and stuttering**.

2. L3 Cache vs DRAM

L3 Cache:

- Smaller capacity.
- Much faster than DRAM.

- Reduces CPU memory-access latency.
- Useful for frequently reused data.

DRAM:

- Much larger capacity.
- Higher latency than L3.
- Provides the working memory required for large game assets.
- High-bandwidth DRAM can transfer large amounts of data quickly.

3. Proposed Memory Hierarchy

CPU/GPU → L1 → L2 → L3 → High-bandwidth DRAM → SSD

4. Proposed Enhancements

1. Increase the effective L3 cache capacity.
2. Use high-bandwidth DRAM with multiple memory channels.
3. Use prefetching to load scene data before it is required.
4. Use fast SSD storage for large game assets.
5. Stream game assets from SSD to DRAM.
6. Compress textures and other assets.
7. Use efficient DMA/direct transfers where possible.

5. Analysis

L3 cache mainly helps to **reduce latency** for frequently reused data.

DRAM mainly provides the **capacity and bandwidth** required for large scenes.

Therefore:

L3 → Low latency

DRAM → High capacity and high bandwidth

SSD → Large permanent game storage

A combination of these levels prevents the GPU from waiting unnecessarily for game assets.

6. Expected Improvement

With:

Fast L3 cache + high-bandwidth DRAM + fast SSD + prefetching

the console can load large scenes more efficiently.

This can reduce:

- Memory stalls
- Loading delays
- GPU waiting time
- Frame drops

and improve:

- Frame consistency
- Throughput
- Overall gaming experience

7. Conclusion

L3 cache and DRAM have different but complementary roles. L3 reduces access latency for frequently used data, while DRAM provides the bandwidth and capacity required for large game scenes. Therefore, a hierarchy containing **L1/L2/L3 cache, high-bandwidth DRAM and fast SSD storage**, supported by prefetching and efficient data transfer, can reduce frame drops and improve gaming performance.

Final Memory Hierarchy Summary

Application	Fast Memory	Main Memory	Advanced/Storage Memory
Smartphone	L1/L2/L3	DRAM	Flash
Cloud Server	L1/L2/L3	DRAM	NVMe SSD
Autonomous Vehicle	SRAM/Cache	DRAM	MRAM
Edge AI	SRAM/Cache	DRAM	RRAM + NAND Flash
Gaming Console	L1/L2/L3	High-bandwidth DRAM	SSD